

CHAPTER 4

Results

This chapter reports empirical outcomes from the three benchmark suites defined in Section 3.8. All runs used GLM 4.7 on frozen Kaggle notebook snapshots with live [API](#) inference. Tables summarise mode-level behaviour; narrative interpretation follows in Chapter 5.

4.1 Experimental Setup

Each suite comprised four scenario tests executed through dedicated runners (`run_agent_tests.py`, `run_ask_tests.py`, and `run_code_tests.py`). Primary notebooks were Pakistan housing (kernel 113620421) and testing-ol (kernel 112732919). Ask and Code modes produced streamed natural-language replies; Agentic mode exercised the mandatory two-phase protocol described in Section 3.5.2.

4.2 Agentic Mode Results

Agentic evaluation focused on **Test 1 (Complete ML Pipeline)** as the primary evidence for tool orchestration. The agent was prompted to create a full machine-learning workflow on the Pakistan housing snapshot.

In this run the model consumed **two API calls**. Round 0 dispatched read-only notebook queries. Round 1 dispatched an implement batch of **37 tools**: twenty-four each of `insert_cell`, `edit_cell_by_index`, and `run_cell`, plus two read operations. Host-side batch reordering preserved writes-before-runs ordering consistent with the execution contract in Section 3.5.2. Total wall time was approximately 9.8 s with zero repair rounds recorded in the harness trace.